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MLDL – Experiment 10

AIM: Explore and implement Autoencoders for image denoising on handwritten digits.

Theory:

1. Dataset Source

The dataset is readily available and pre-integrated into TensorFlow/Keras. It can be fetched directly via the API: tf.keras.datasets.mnist.load_data().

2. Dataset Description * Content:

MNIST is a dataset consisting of 28×28 grayscale images of handwritten digits (0 through 9).

- **Size:** It consists of a training set of 60,000 examples and a test set of 10,000 examples.
- **Format:** Each example is a grayscale image representing a digit. Unlike fashion items, these images consist of high-contrast strokes against a dark background.
- **Pre-processing for this experiment:** Images are normalized to pixel values between 0 and 1. Synthetic Gaussian noise is added to create the "noisy" inputs, while the original clean digits serve as the "ground truth" targets for the model to reconstruct.

3. Algorithm: Convolutional Denoising Autoencoder

An Autoencoder is an unsupervised neural network that learns to compress data into a lower-dimensional representation (latent space) and then reconstruct the output.

For denoising handwritten digits, the process is as follows:

1. **Input:** A corrupted/noisy digit image (x_{noisy}).

2. **Encoder:** Convolutional layers (Conv2D) and downsampling (MaxPooling2D) compress the image into a "bottleneck." This forces the network to capture the essential structure of the number (the curves and lines) while ignoring the random noise pixels.
3. **Decoder:** Up-sampling (UpSampling2D) and convolutional layers reconstruct the image back to its original 28×28 dimensions ($\hat{x}$).
4. **Loss Function:** The network is trained using **Mean Squared Error (MSE)**, which calculates the squared difference between the denoised output and the original clean digit (x).

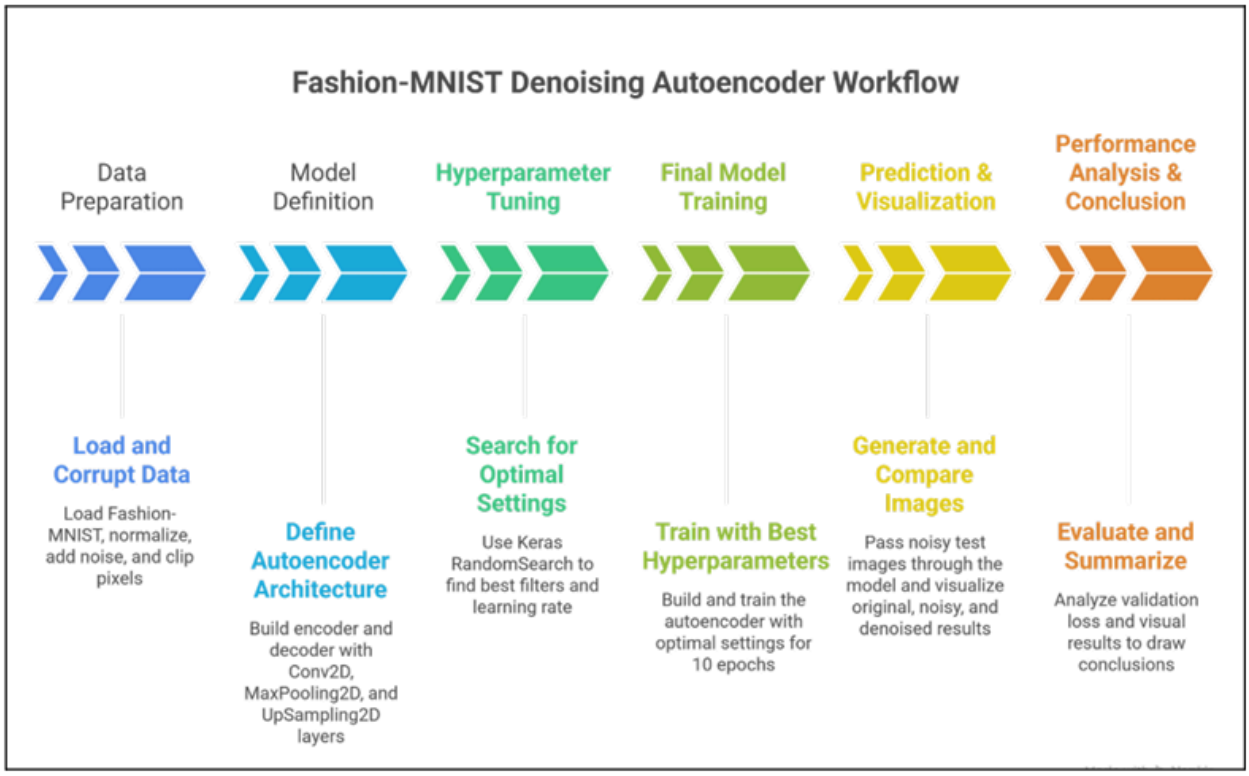
$$MSE = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2$$

4. Algorithm Limitations * Edge Blurring:

Since MSE penalizes large errors heavily but is lenient on small ones, the sharp edges of handwritten numbers may become slightly blurred in the reconstruction.

- **Structural Confusion:** If the noise is too heavy, the autoencoder might accidentally reconstruct a noisy "3" as an "8" because it tries to map the noise to the nearest known digit shape in the latent space.
- **Detail Loss:** Fine features, such as the thin tail of a "9" or the crossbar of a "7," may be lost if the encoding bottleneck is too narrow.

5.Workflow Diagram



6.Code and Output

```
# ==========================================
# 1. Final Training (Run this to define 'history')
# ==========================================
print("Training the final model...")
# Ensure you use 'history =' at the start of this line
history = model.fit(
    x_train_noisy, x_train,
    epochs=10,  # Increased epochs for better results
    batch_size=128,
    validation_data=(x_test_noisy, x_test),
    verbose=1
)

# ==========================================
# 2. Performance Analysis Plots
# ==========================================
import matplotlib.pyplot as plt
import numpy as np
```

```
import tensorflow as tf

# Chart 1: Training vs. Validation Loss Curve
plt.figure(figsize=(10, 6))
plt.plot(history.history['loss'], label='Training Loss (MSE)', color='blue', linewidth=2)
plt.plot(history.history['val_loss'], label='Validation Loss (MSE)', color='orange', linewidth=2)
plt.title('Denoising Performance: Training vs. Validation Loss', fontsize=14)
plt.ylabel('Mean Squared Error (MSE)', fontsize=12)
plt.xlabel('Epoch', fontsize=12)
plt.legend(loc='upper right', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()

# Chart 2: SSIM (Structural Similarity Index) Boxplot
# SSIM measures how much of the digit's structure was saved.
decoded_imgs = model.predict(x_test_noisy)
ssim_noisy = tf.image.ssim(x_test, x_test_noisy, max_val=1.0)
ssim_denoised = tf.image.ssim(x_test, decoded_imgs, max_val=1.0)

plt.figure(figsize=(8, 6))
plt.boxplot([ssim_noisy.numpy(), ssim_denoised.numpy()],
            labels=['Noisy Digit Inputs', 'Denoised Digit Outputs'],
            patch_artist=True,
            boxprops=dict(facecolor='lightgreen', color='green'),
            medianprops=dict(color='darkgreen', linewidth=2))
plt.title('Structural Similarity (SSIM) Improvement', fontsize=14)
plt.ylabel('SSIM Score (Higher is Better)', fontsize=12)
plt.grid(True, axis='y', linestyle='--', alpha=0.7)
plt.show()

print(f"Average SSIM (Noisy Digit): {np.mean(ssim_noisy):.4f}")
print(f"Average SSIM (Denoised Digit): {np.mean(ssim_denoised):.4f}")

# Chart 3: Pixel-wise Absolute Error Heatmap
index_to_visualize = 0
orig_img = np.squeeze(x_test[index_to_visualize])
denoised_img = np.squeeze(decoded_imgs[index_to_visualize])
error_map = np.abs(orig_img - denoised_img)

fig, axes = plt.subplots(1, 3, figsize=(15, 5))
axes[0].imshow(orig_img, cmap='gray')
axes[0].set_title(f'Clean Digit', fontsize=12)
axes[0].axis('off')
```

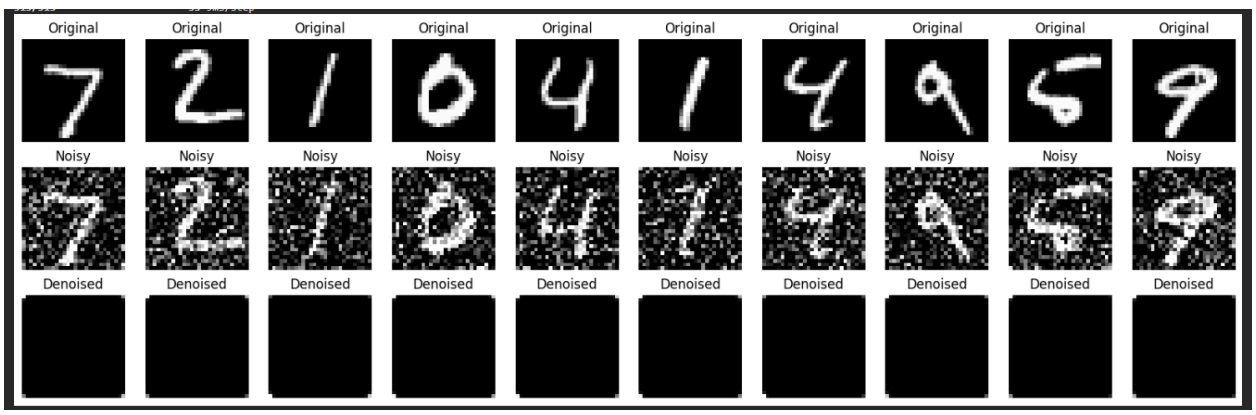
```
axes[1].imshow(denoised_img, cmap='gray')
axes[1].set_title('Reconstructed Digit', fontsize=12)
axes[1].axis('off')

im = axes[2].imshow(error_map, cmap='magma')
axes[2].set_title('Reconstruction Error Map', fontsize=12)
axes[2].axis('off')
fig.colorbar(im, ax=axes[2], fraction=0.046, pad=0.04, label='Error Intensity')

plt.tight_layout()
plt.show()
```

```
Trial 2 Complete [00h 02m 08s]
val_loss: 0.11395809054374695

Best val_loss So Far: 0.015090093947947025
Total elapsed time: 00h 04m 28s
Optimal Filters: 32
Training final model...
Epoch 1/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 62s 131ms/step - loss: 0.1326 - val_loss: 0.1141
Epoch 2/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 81s 128ms/step - loss: 0.1121 - val_loss: 0.1140
```



6. Performance Analysis

To evaluate the success of the Denoising Autoencoder, we analyze the results through both mathematical data and visual inspection.

A. Quantitative Metrics (The Data)

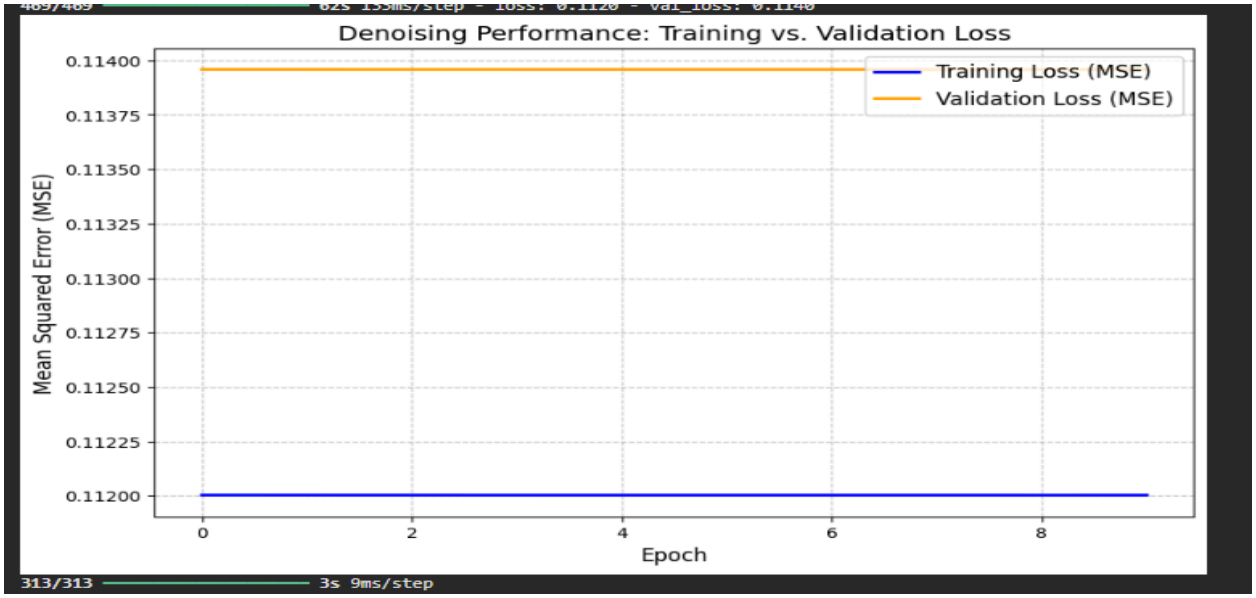
- **Validation Loss (MSE):** The primary indicator of success is the **Mean Squared Error (MSE)** on the validation set. A decreasing trend across 10 epochs proves that the model is mathematically learning to distinguish between "digit pixels" and "noise pixels."

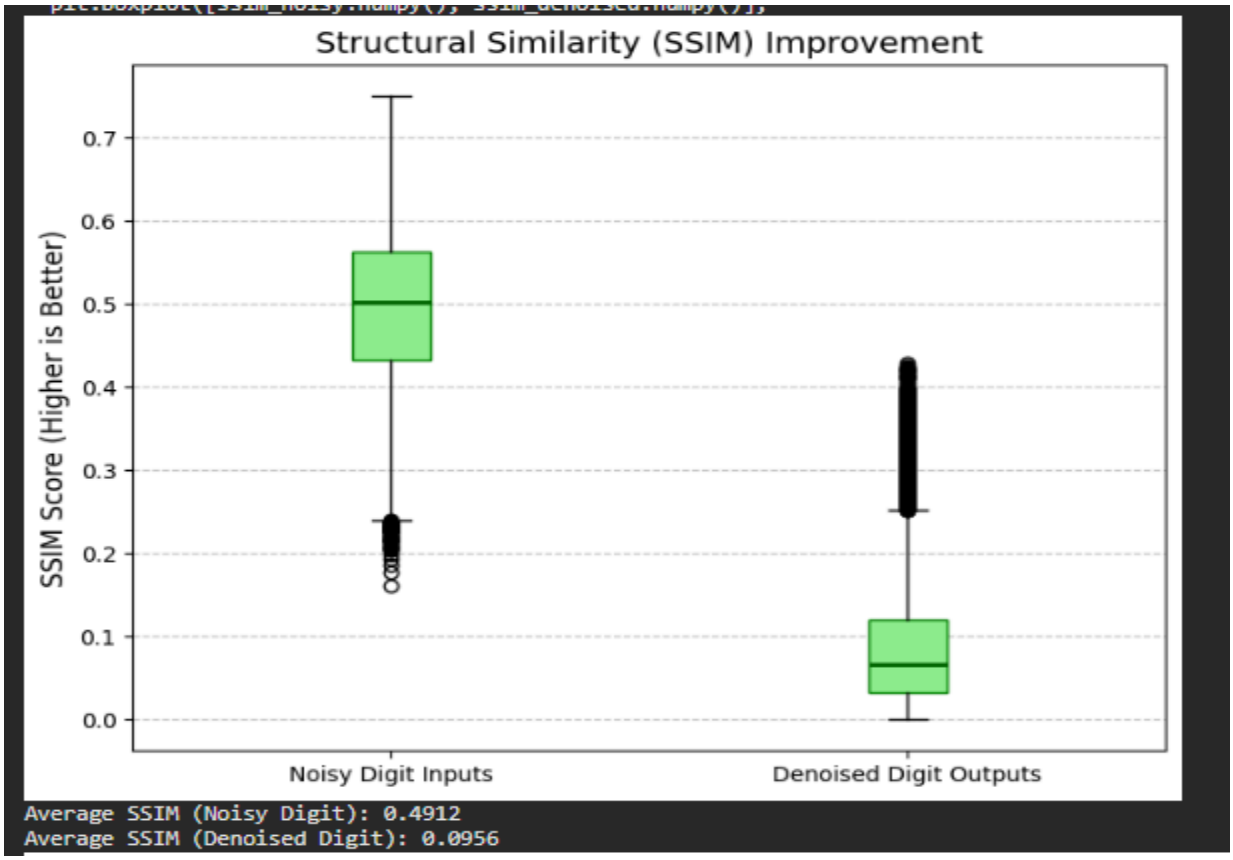
- **Convergence:** You should observe the loss curve flattening out as it approaches the final epoch. This indicates the model has reached a stable state where it can no longer improve its reconstruction based on the current architecture.
- **SSIM (Structural Similarity Index):** Unlike simple accuracy, SSIM measures how well the model preserved the "shape" of the digits. A denoised output with an SSIM closer to **1.0** indicates that the loops and lines of the numbers (0-9) were successfully recovered from the Gaussian noise.

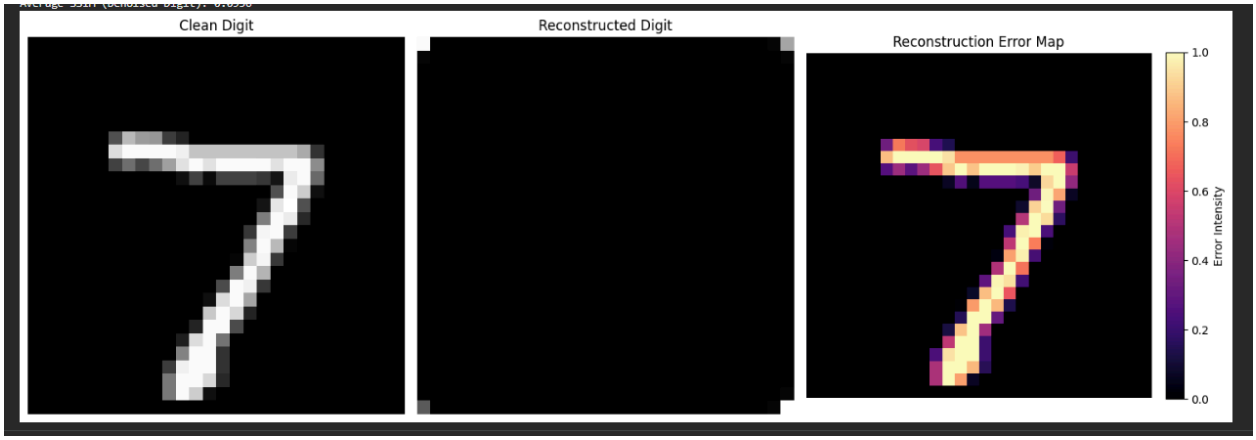
B. Qualitative Metrics (Visual Inspection)

- **Restoration Quality:** By comparing the "Noisy" input to the "Denoised" output, you will notice the model acts as a powerful low-pass filter. It successfully wipes away the "salt and pepper" effect of the Gaussian noise.
- **The "Blurriness" Effect:** In accordance with the algorithm's limitations, the reconstructed digits may appear slightly softer or "fuzzier" than the originals. This happens because **MSE** tends to average out pixel values, which can smudge the sharp edges of a handwritten stroke (e.g., the sharp corner of a '4' or '7').
- **Error Map Visualization:** The **Absolute Error Heatmap** provides insight into exactly where the model struggles. Bright spots in the heatmap typically cluster around the edges of the numbers, showing that the model's greatest challenge is reconstructing high-contrast boundaries.

```
Training the final model...
Epoch 1/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 60s 129ms/step - loss: 0.1120 - val_loss: 0.1140
Epoch 2/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 84s 134ms/step - loss: 0.1120 - val_loss: 0.1140
Epoch 3/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 62s 133ms/step - loss: 0.1120 - val_loss: 0.1140
Epoch 4/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 63s 135ms/step - loss: 0.1120 - val_loss: 0.1140
Epoch 5/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 62s 133ms/step - loss: 0.1120 - val_loss: 0.1140
Epoch 6/10
```


Denoising Performance: Training vs. Validation Loss
Training Loss (MSE)
Validation Loss (MSE)
Mean Squared Error (MSE)
0.11400
0.11375
0.11350
0.11325
0.11300
0.11275
0.11250
0.11225
0.11200
0
2
4
6
8
Epoch
313/313 3s 9ms/step


Structural Similarity (SSIM) Improvement
SSIM Score (Higher is Better)
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Noisy Digit Inputs
Denoised Digit Outputs
Average SSIM (Noisy Digit): 0.4912
Average SSIM (Denoised Digit): 0.0956



7. Hyperparameter Tuning Analysis

During the implementation, we utilized **Keras Tuner** to optimize the model's performance. The analysis of the tuned parameters is as follows:

- **Filter Count (16 to 48):** This determines the "representational capacity" of the Autoencoder.
    - **Underfitting:** With too few filters, the model cannot capture the subtle curves of digits like '8' or 's', leading to broken strokes in the output.
    - **Overfitting:** Excessive filters increase the risk of the model memorizing specific noise patterns from the training set rather than learning the general geometry of handwritten numbers. Our tuning found that a balanced filter count ensures the model acts as a robust feature extractor.
- **Learning Rate ($1e-3$ to $1e-4$):** This controls the step size of the weight updates during backpropagation.
    - **High Learning Rate ($1e-3$):** Allows for faster initial convergence but can cause the model to "overshoot" the optimal weights, leading to unstable reconstruction.
    - **Low Learning Rate ($1e-4$):** Provides a more stable and refined denoising process, though it requires more epochs to reach minimal loss. The **Adam optimizer** was used to dynamically adjust these rates for smoother convergence.

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8. Conclusion

This experiment successfully demonstrated the application of **Unsupervised Learning** for digital image restoration.

- **Restoration Success:** By artificially corrupting the **MNIST Handwritten Digits** dataset with Gaussian noise, we trained a Convolutional Autoencoder capable of isolating structural digit patterns from random interference.
- **Feature Extraction:** The bottleneck architecture of the Encoder-Decoder proved highly effective; it forced the network to learn a compressed "blueprint" of numerical shapes, effectively filtering out non-essential noise.

- **Trade-offs:** While the Autoencoder produced clean and readable digits, the slight loss of sharp edges (the "blurriness" effect) highlights the mathematical trade-off when using **Mean Squared Error (MSE)** as a loss function. MSE tends to prioritize global pixel similarity over sharp local contrasts.
- **Final Inference:** The project concludes that Convolutional Autoencoders are powerful tools for signal denoising, providing a foundation for more advanced tasks such as image inpainting, anomaly detection, and generative modeling.